Genome-wide polymorphism and gene expression analysis of the citrus mutant 'Yein-early'

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Introduction

Yein-early is a gamma-irradiated mutant of Citrus unshiu that exhibits two stable traits distinct from the wild type (WT): a deeper red peel color and narrow, curled leaves. Fruit quality analysis showed that Yein-early has higher firmness and sugar content, while fruit size, acidity, and peel thickness were comparable to the WT. To uncover the genetic basis of these traits, we conducted whole-genome sequencing. The WT sequence data were sourced from our previous work (NCBI: PRJNA745525). Reads were mapped to the C. unshiu Marc. Miyagawa-wase reference genome (CUMW_v1.0; 359.7 Mb), producing 55,897 Mbp and 56,418 Mbp of clean reads for Yein-early and WT, with mapping rates of 84.8% and 86.9%, respectively. Comparative genomic analysis revealed extensive polymorphisms in Yein-early: 650,257 SNPs and 105,817 InDels, including 72,155 homozygous SNPs and 4,683 homozygous InDels. Most variants occurred in gene regions (26,339 genes for SNPs; 14,889 for InDels), indicating broad genomic perturbation. SNP polymorphisms were markedly more abundant than InDels, and Yein-early displayed a large number of fixed mutations, suggesting radiation-induced genomic restructuring. Transcriptome profiling of immature fruit identified differentially expressed genes. Up-regulated genes were associated with metabolism, transport, and photosynthetic processes, while genes linked to ribosomal function and redox balance were down-regulated. These multi-omic data connect phenotype, genomic variation, and transcriptional regulation, providing insight into the mechanisms underlying the peel pigmentation and altered leaf morphology of Yein-early.

Results

[Sampling stages of leaf, flower, and fruit development in *Citrus unshiu*]

Fig 1. Leaf, flower, and five developmental stages of *Citrus unshiu* fruit used for morphological observation and gene expression analysis.

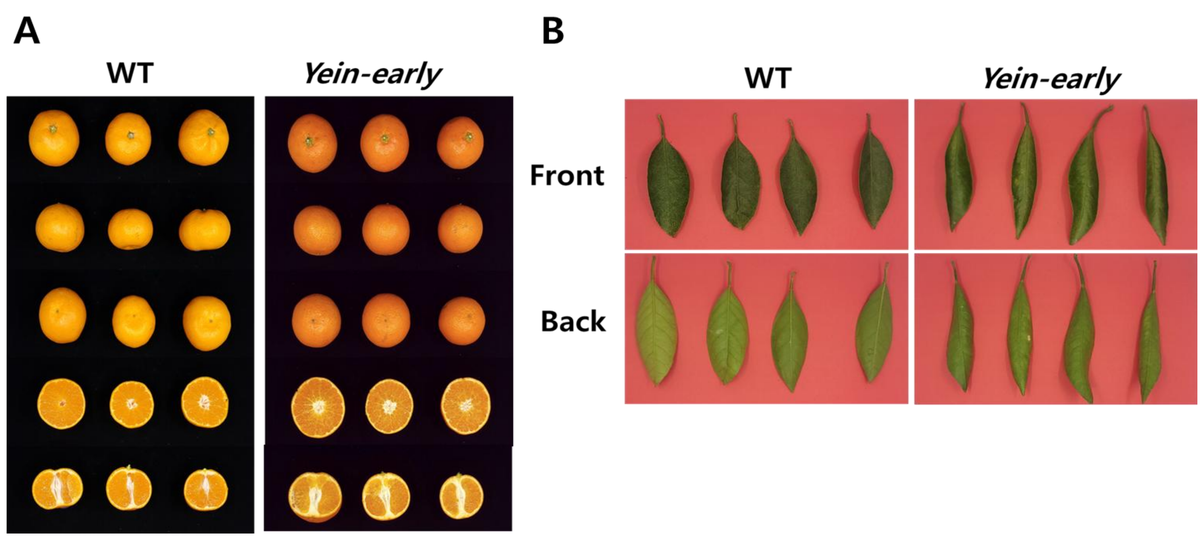

Fig 2. (A) Morphological comparison of fruits and (B) leaves between 'Yein-early' and Wild type(WT)

Table 1. Comparative analysis of Wild type(WT) control and 'Yein-early' fruit

	Year	Horizontal (mm)	Vertical (mm)	Weight (g)	Peel thickness (mm)	Hardness
Wild type	2023	48.63±2.79	64.8±5.04	111.4±15.98	2.59±0.95	870.4±259.4
	2024	59.1±7.2	72.358±5.7	167.08±38.1	3.29±1.05	1030.5±430.9
Yein-early	2023	50.28±3.22	62.19±4,71	96.92±14.95	3.33±0.61	1151.6±343.4
	2024	50.86±3.8	60.03±6.22	101.4±20.1	2.69±0.86	1241.1±342.3

	Year	Hunter color value			Sugar (Brix)	Acidity (wt%)
		L	a	b		
Wild type	2023	58.48±1.9	22.8±2.15	35.74±1	8.6±1.05	0.51±0.07
	2024	67.5±3.36	18.21±8.02	40.5±2.7	7.87±0.56	0.71±0.12
Yein-early	2023	56.14±1.45	34.83±1.73	33.06±1.04	10.14±1.16	0.76±0.09
	2024	68.34±4.79	34.57±3.06	59.1±15.7	8.57±0.66	0.75±0.08

	Longitudinal Length	Transverse Length	Longitudinal / Transverse	Curling	Surface Undulation	Edge Waviness	Edge Indentations	Shape of the Apex	Groove at the Tip
WT	95.44±7.58	42.50±4.52	2.26±0.16	weak	weak	weak	blunt saw blade shape	sharp	no
Yein-early	108.77±7.14	35.02±4.69	3.15±0.44	strong	Weak	weak	blunt saw blade shape	sharp	no

[Genome-wide polymorphism analysis]

Table 2. Statistics of polymorphic SNPs and In/Dels between the WT and Yein-early

Sample		No. of Total polymorphic[1]	No. of polymorphic	
			Homo[2]	Hetero[3]
WT vs Yein-early	SNP	650,257	72,155	578,102
	In/Del	105,817	4,683	101,134

[1] SNP and In/Del loci showing differences in nucleotide sequences between comparative samples
[2] When the nucleotide sequences of both samples are homozygous
[3] When the nucleotide sequence is homozygous in one sample and heterozygous in the other sample

Table 3. Statistics of Gene Ontology (GO) annotations for polymorphic SNPs and InDels between the WT and Yein-early

Sample		Polymorphic	Genes[1]	GO genes[2]
WT vs Yein-early	SNP	650,257	26,339	23,508
	In/Del	105,817	14,889	13,685

[1] Number of genes corresponding to SNP or In/Del loci
[2] Number of genes annotated in the GO database among the genes corresponding to SNP or InDel loci (e-value ≤1e-10, best hits)

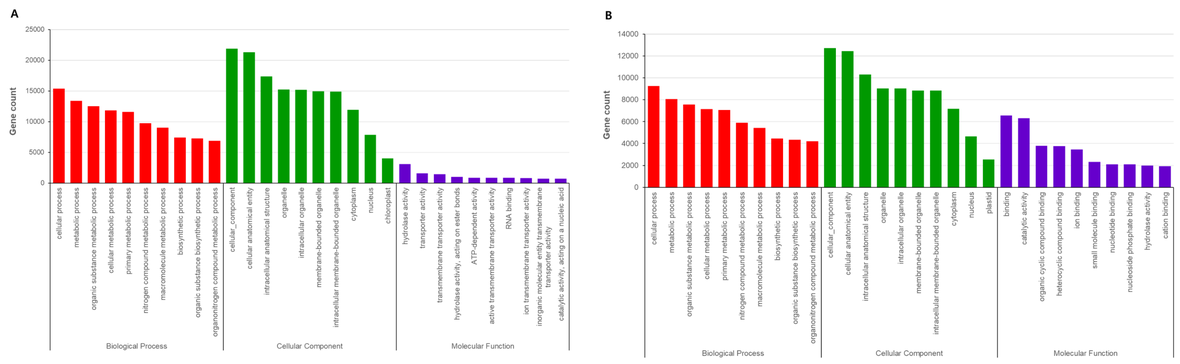

Fig 3. GO enrichment of SNP- and InDel-containing genes (WT vs. Yein-early).
(A) Top 10 GO terms for SNPs. (B) Top 10 GO terms for InDels.

[Transcriptome-based expression analysis]

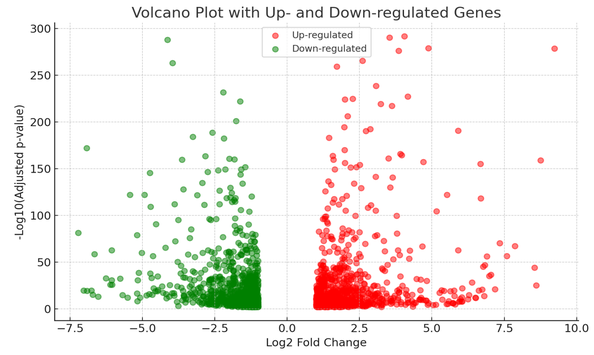

Fig 4. Volcano Plot of up- and down-regulated genes. Green and red bars represent a down- and up-regulated genes

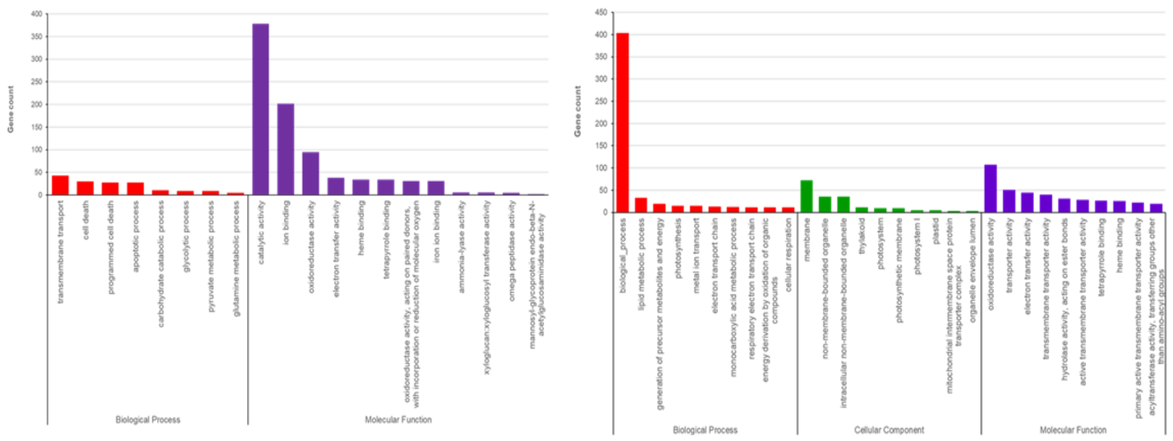

Fig. 5. GO functional enrichment analysis of DEGs (Yein-early vs WT), showing pathways linked to pigmentation and metabolism.

Acknowledgements

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